

Dichotomizing Temporarily Bound Objects Using Capture States

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Introduction

Temporarily Bound Objects (TBOs) are Near-Earth Objects (NEO) **temporarily captured** by the Earth-Moon System (EMS). With TBOs being small and their observations sparse, establishing a mapping between their arrival conditions and their taxa would reduce the need for observational and computational resources, enhancing rapid detection of mission-viable targets.

A TBO is an NEO with:

1) Geocentric Keplerian energy < 0

2) Geocentric distance ≤ 3 Earth Hill radius (EH)

3) Closest approach to Earth ≤ 1 EH

If it also satisfies:

$\geq 360^\circ$ cumulative absolute change in geocentric synodic longitude

It is a **Temporarily Captured Orbiter (TCO)**; otherwise, a **Temporarily Captured Flyby (TCF)** [1].

A TBO's entire trajectory through the EMS under the gravitational influence of all major bodies, the Moon and Pluto is **deterministic**, so the entire future trajectory is fixed by the state of capture; but it is also **chaotic**, so the map from capture state to the taxa is correspondingly complex.

A developed machine learning model shows to recover 79.50% of the taxa without explicitly representing the underlying physics, using only the geocentric Keplerian orbital elements at capture, with RAAN expressed in the Sun-Earth geocentric synodic frame.

Classification

Six classifiers are implemented, two of which are generative, and four discriminative. To classify an object x as class y , a **generative classifier attempts to build the joint distribution model $p(x, y)$** , whereas a **discriminative classifier merely seeks to learn $p(x|y)$** , or in some cases the decision boundary directly.

Generative classifiers:

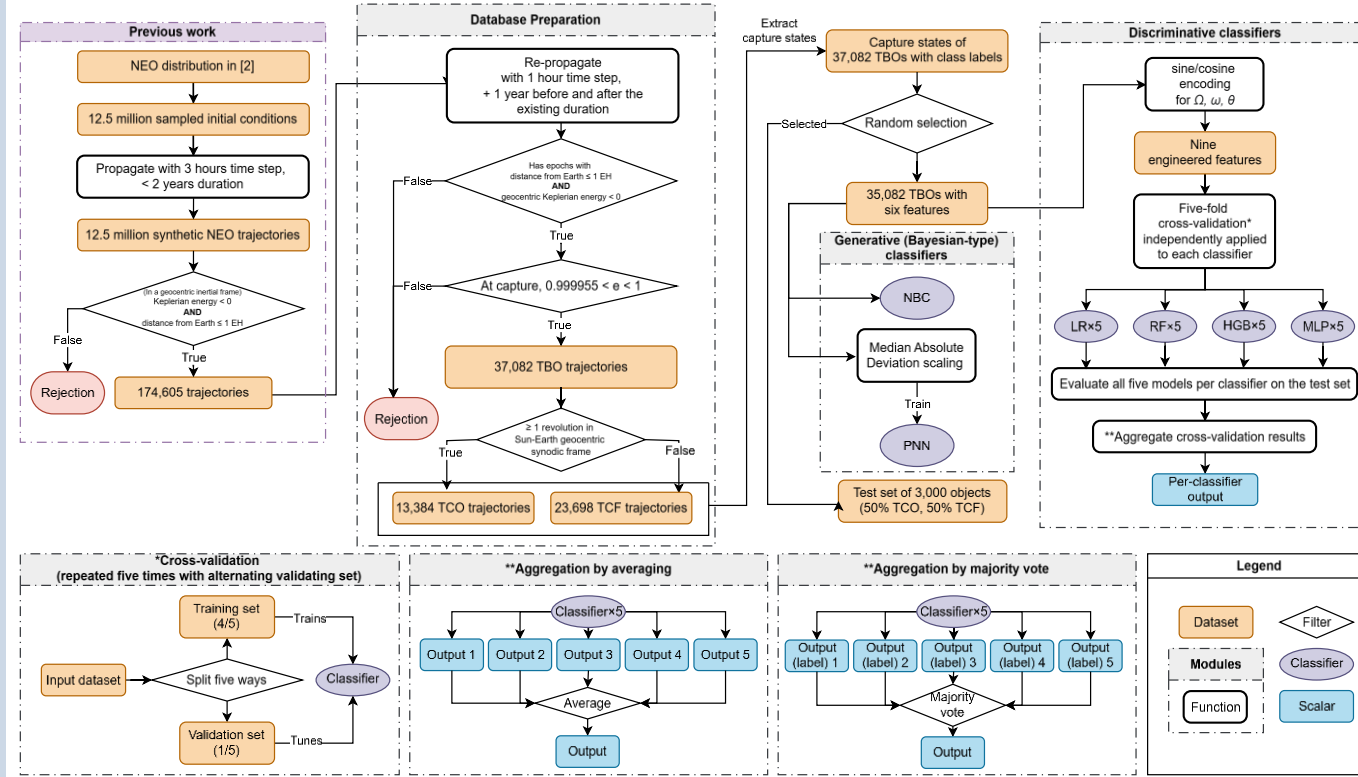
- Naive Bayes Classifier (NBC)
- Probabilistic Neural Network (PNN)

Discriminative classifiers:

- Logistic Regression (LR)
- Random Forest (RF)
- Histogram Gradient Boost (HGB)
- Multilayer Preceptron (MLP)

A balanced test set of 3,000 objects was selected for final evaluation of classifier performance. The remaining data underwent cross-validation with the data split into five equal-sized segments. Each segment serves once as validation data while the other four as training data. The resulting five models are then combined through either **majority voting** or **averaging** aggregation schemes.

Classification Procedures



Metrics

	Predicted TCF	Predicted TCO	Actual TCF	True Positive (TP)	False Negative (FN)	Actual TCO	False Positive (FP)	True Negative (TN)
Accuracy	$\frac{TP + TN}{TN + FP + TP + FN}$							
Precision	$\frac{TP}{TP + FP}$							
Recall	$\frac{TP}{TP + FN}$							
Specificity	$\frac{TN}{TN + FP}$							
NPV	$\frac{TN}{TN + FN}$							
f1-score	$2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$							

Discussion

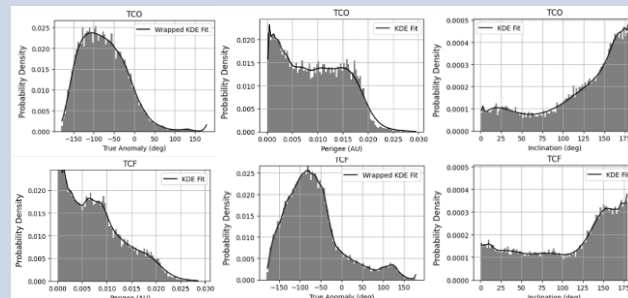
- The capture state of a TBO carries substantial predictive information about its taxa.
- The MLP classifier reaches **79.50%** accuracy on a balanced test set. Its affine-plus-ReLU compositions enable it to directly learn nonlinear relationships rather than approximating them through piecewise constant approximations.
- Adjusting for the TCF dominance in the synthetic database, MLP has an expected accuracy of 82.4%, with a negative predictive value for TCO of 79.3%, as opposed to the natural fraction of 36.1%. This translates to a 2.2x enrichment of TCOs in the follow-up queue. This enrichment is directly convertible into observing-time savings.

Future Research

- Introduce realistic orbital determination uncertainties through the state and epoch at capture to test whether the classification remains robust.
- Supply the classifier with physically motivated derived features, such as the osculating Sun-Earth geocentric semi-major axis, Jacobi constant, and the Earth-Moon geometry at capture.
- Build the classifier to predict the cumulative synodic angle, with TCO/TCF label recovered by thresholding. The objects on the margin of the one-revolution boundary will therefore contribute their true information content.
- Admit objects with repeated captures as a distinct third class.

References

- [1] Sean Wolfe and M. Reza Emami. Temporary Captures in Earth-Moon System: A Taxonomy Design using Machine Learning. The Journal of the Astronautical Sciences, 71(6):52, November 2024. ISSN 2195-0571. doi: 10.1007/s40295-024-00473-4. URL <https://doi.org/10.1007/s40295-024-00473-4>.
- [2] Grigori Fedorets, Mikael Granvik, and Robert Jedicke. Orbit and size distributions for asteroids temporarily captured by the earth-moon system. Icarus, 285:83–94, 2017. ISSN 0019-1035. doi: 10.1016/j.icarus.2016.12.022. URL <https://www.sciencedirect.com/science/article/pii/S0019103516306480>



Model	Aggregation Method	Threshold	Accuracy	Precision	Recall	Specificity	NPV	F1
Naive Bayes Classifier	N/A	N/A	0.6170	0.6139	0.6307	0.6033	0.6203	0.6222
Probabilistic Neural Network	N/A	N/A	0.6894	0.6768	0.7247	0.6540	0.7037	0.6999
Logistic Regression	Majority Vote	0.210	0.5000	0.5000	1.0000	0.0000	0.0000	0.6667
	Averaging	0.210	0.5000	0.5000	1.0000	0.0000	0.0000	0.6667
Random Forest	Majority Vote	0.515	0.7530	0.7024	0.8780	0.6280	0.8373	0.7804
	Averaging	0.515	0.7464	0.6956	0.8760	0.6167	0.8326	0.7754
Histogram Gradient Boost	Majority Vote	0.535	0.7190	0.6732	0.8513	0.5867	0.7978	0.7518
	Averaging	0.535	0.7210	0.6744	0.8547	0.5873	0.8016	0.7539
Multilayer Perceptron	Majority Vote	0.495	0.7927	0.7417	0.8980	0.6873	0.8708	0.8124
	Averaging	0.495	0.7950	0.7446	0.8980	0.6920	0.8715	0.8141